

Client/ClientUI.py

```
# client_ui.py
# í .° ...ì .ì ,í , UI Εμ<ì - °"□°

import sys
import os
sys.path.append(os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
from tkinter import *
from tkinter import ttk, scrolledtext
from tkinter.font import Font
from env_utils import DEFAULT_HOST, DEFAULT_PORT

class ClientUI:
    """í .° ...ì .ì ,í , UI Εμ<ì - í .° ì /"""

    def __init__(self, root):
        self.root = root
        self.root.title("AI í ì μ ì § ì ° Εμ<")
        self.root.geometry("1400x820")
        self.root.minsize(1300, 780)
        self.root.configure(bg='#f5f5f5')

        # í ° í , ì /ì
        self.title_font = Font(family="°§ ì Ε† ° ", size=18, weight="bold")
        self.heading_font = Font(family="°§ ì Ε† ° ", size=12, weight="bold")
        self.normal_font = Font(family="°§ ì Ε† ° ", size=10)

        # °ì Ε•,ì , ì í
        self.is_logged_in = False
        self.current_user_email = None
        self.current_username = None

        # UI Εμ<ì -
        self.create_widgets()

    def create_widgets(self):
        """ì ì fl ì ì -"""
        # °' ì , ì »í ì .°
        main_container = Frame(self.root, bg='#f5f5f5')
        main_container.pack(fill=BOTH, expand=True)

        # ì <ì .° °°
        self.create_sidebar(main_container)

        # °' ì , ì % í ì , ì ì >
        self.create_main_content(main_container)

    def create_sidebar(self, parent):
        """ì <ì .° °° ì ì -"""
        sidebar = Frame(parent, bg='ffffff', width=300)
        sidebar.pack(side=LEFT, fill=Y, padx=0, pady=0)
        sidebar.pack_propagate(False)

        # ì <ì .° °° í /°
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ClientFrame(pydebar, bg='#333333', height=80)

header.pack(fill=X)

Label(header, text="ð AI í ì μ ì § ì ° Æ μ <", font=self.title_font,
bg='#333333', fg='white').pack(pady=20)

ì ° † ì ° Æ † ° ì / ì (ì ° ì ° Æ † °, UI ° ì ° Æ, ° ì ì ')

server_frame = LabelFrame(sidebar, text="ð ì ° † ì ° Æ † °", font=self.heading_font,
bg='ffffff', fg='#333', padx=15, pady=10)

server_frame.pack(fill=X, padx=10, pady=10)

ì ° † ì ° ‡ . í ì (ì ° Æ, ° ì ì ')

Label(server_frame, text="ì ° † ì £... ì :", font=self.normal_font, bg='ffffff').pack(anchor=

self.server_host_label = Label(server_frame, text=DEFAULT_HOST,
font=self.normal_font, bg='ffffff', fg='#333')

self.server_host_label.pack(anchor=W, pady=2)

Label(server_frame, text="í < í ,:", font=self.normal_font, bg='ffffff').pack(anchor=W, pady

self.server_port_label = Label(server_frame, text=str(DEFAULT_PORT),
font=self.normal_font, bg='ffffff', fg='#333')

self.server_port_label.pack(anchor=W, pady=2)

self.server_status_label = Label(server_frame, text="ì ° Æ † ° ì í : ì ° Æ † ° ì / ...",
font=self.normal_font, bg='ffffff', fg='#666')

self.server_status_label.pack(pady=10)

AI ì ì ì ° ' ° .

action_frame = LabelFrame(sidebar, text="ð fl AI ì ì ì ° ' ° .", font=self.heading_font,
bg='ffffff', fg='#333', padx=15, pady=10)

action_frame.pack(fill=X, padx=10, pady=10)

self.convert_btn = Button(action_frame, text="ð ì , ° „ ì í ì ... ° ‡ í ",
bg='#333333', fg='white', font=self.normal_font, relief=FLAT, pady=10)

self.convert_btn.pack(fill=X)

í . ° Æ . ° f <

folder_frame = LabelFrame(sidebar, text="ð í . ° Æ . ° f <", font=self.heading_font,
bg='ffffff', fg='#333', padx=15, pady=10)

folder_frame.pack(fill=X, padx=10, pady=10)

self.folder_btn = Button(folder_frame, text="ð í . ° Æ . ° f <",
bg='#333333', fg='white', font=self.normal_font, relief=FLAT, pady=10)

self.folder_btn.pack(fill=X)

í ° ì ì í , Æ, ° ° ì

history_frame = LabelFrame(sidebar, text="ð í ° ì ì í , Æ, ° ° ì ", font=self.heading_font,
bg='ffffff', fg='#333', padx=15, pady=10)

history_frame.pack(fill=BOTH, expand=True, padx=10, pady=10)

í , ° f < ° ° ° ì í ° ì ì í , í ì

self.project_tree = ttk.Treeview(history_frame, show='tree')

self.project_tree.pack(fill=BOTH, expand=True)

° £ ¢ í , ° , ° (° ‡ í ° í ì ...)

self.project_root = self.project_tree.insert('', 'end', text='° ‡ í ° í ì ...', open=True)

def create_main_content(self, parent):

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main_content = Frame(parent, bg='#ffffff')
main_content.pack(side=LEFT, fill=BOTH, expand=True, padx=30, pady=30)
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# °<,ì °°ì .ì»í à .°
viewer_frame = Frame(main_content, bg='#ffffff')
viewer_frame.pack(fill=BOTH, expand=True)
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# °<,ì í /°
self.document_header = Frame(viewer_frame, bg='#ffffff')
self.document_header.pack(fill=X, pady=(0, 20))
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```
self.document_title = Label(self.document_header, text="AI í à μ ì § ì ° æμ<",
                             font=self.title_font, bg='#ffffff', fg='#333')
self.document_title.pack(anchor=W)
```

```
# ì ,° -ì / °† í ...
self.index_frame = Frame(viewer_frame, bg='#ffffff')
self.index_frame.pack(fill=X, pady=(0, 20))
self.index_frame.pack_forget()
```

```
self.summary_btn = Button(self.index_frame, text="ð ì ì % ì °f<",
                           bg='#333333', fg='white', font=self.normal_font, relief=FLAT, padx=5)
self.summary_btn.pack(side=LEFT, padx=5)
```

```
self.quiz_btn = Button(self.index_frame, text="â í .ìf ",
                       bg='#f0f0f0', fg='#666', font=self.normal_font, relief=FLAT, padx=20)
self.quiz_btn.pack(side=LEFT, padx=5)
```

```
# ì% í ì, ì ì > (ì /í <°ì/ æ° ° ¥)
self.content_canvas = Canvas(viewer_frame, bg='#ffffff', highlightthickness=0)
scrollbar = Scrollbar(viewer_frame, orient="vertical", command=self.content_canvas.yview)
self.content_frame = Frame(self.content_canvas, bg='#ffffff')
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self.content_canvas.pack(side=LEFT, fill=BOTH, expand=True)
scrollbar.pack(side=RIGHT, fill=Y)
self.content_canvas.configure(yscrollcommand=scrollbar.set)
```

```
# Canvasì ° °,, ì °§ ì¶° content_frameì . í ì ¥° ° °ì ì /ì
self.content_window = self.content_canvas.create_window((0, 0), window=self.content_frame, a
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```
def _update_content_canvas(event):
    # Canvas ° °,, ì °§ ì¶° content_frame ° °,, ì ì°ì
    canvas_width = event.width
    self.content_canvas.itemconfig(self.content_window, width=canvas_width)
    self.content_canvas.configure(scrollregion=self.content_canvas.bbox("all"))
```

```
self.content_canvas.bind("<Configure>", _update_content_canvas)
self.content_frame.bind("<Configure>", lambda e: self.content_canvas.configure(scrollregion=
```

```
# ì ìì í ...° /
self.create_convert_form()
self.create_folder_form()
self.create_summary_section()
self.create_quiz_section()
```

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```
def create_convert_form(self):
    """
    """
    self.convert_frame = LabelFrame(self.content_frame, text="Convert",
                                     font=self.heading_font, bg='ffffff', padx=25, pady=25)
    self.convert_frame.pack(fill=X, pady=10)
    self.convert_frame.pack_forget()

    Label(self.convert_frame, text="Enter the file name to convert",
          font=self.normal_font, bg='ffffff', fg='#666', wraplength=800).pack(anchor=W, pady=10)

    file_frame = Frame(self.convert_frame, bg='ffffff')
    file_frame.pack(fill=X, pady=10)

    self.audio_file_label = Label(file_frame, text="Audio file",
                                   font=self.normal_font, bg='ffffff', fg='#666')
    self.audio_file_label.pack(anchor=W, pady=5)

    self.select_file_btn = Button(file_frame, text="Select file",
                                   bg='#2196F3', fg='white', font=self.normal_font, relief=FLAT, padx=15, pady=5)
    self.select_file_btn.pack(side=LEFT, padx=5)

    self.convert_btn_main = Button(self.convert_frame, text="Convert",
                                   bg='#333333', fg='white', font=self.normal_font, relief=FLAT, padx=20, pady=10)
    self.convert_btn_main.pack(pady=10)

    self.convert_result_label = Label(self.convert_frame, text="", font=self.normal_font,
                                       bg='ffffff', fg='green', wraplength=800, justify=LEFT)
    self.convert_result_label.pack(anchor=W, pady=10)

def create_folder_form(self):
    """
    """
    self.folder_frame = LabelFrame(self.content_frame, text="Folder",
                                     font=self.heading_font, bg='ffffff', padx=25, pady=25)
    self.folder_frame.pack(fill=X, pady=10)
    self.folder_frame.pack_forget()

    # Create a tab frame
    tab_frame = Frame(self.folder_frame, bg='ffffff')
    tab_frame.pack(fill=X, pady=(0, 20))

    self.folder_tab_add = Button(tab_frame, text="Add",
                                  bg='#333333', fg='white', font=self.normal_font, relief=FLAT,
    self.folder_tab_add.pack(side=LEFT, padx=5)

    self.folder_tab_update = Button(tab_frame, text="Update",
                                      bg='f0f0f0', fg='#666', font=self.normal_font, relief=FLAT,
    self.folder_tab_update.pack(side=LEFT, padx=5)

    self.folder_tab_delete = Button(tab_frame, text="Delete",
                                      bg='f0f0f0', fg='#666', font=self.normal_font, relief=FLAT,
    self.folder_tab_delete.pack(side=LEFT, padx=5)

    # Create a section for adding new files
    self.folder_add_section = Frame(self.folder_frame, bg='ffffff')
```

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```
self.folder_add_section.pack(fill=X, pady=10)

Label(self.folder_add_section, text="í .° ì .°f :", font=self.normal_font, bg='#ffffff').pack()
self.folder_name_add_entry = Entry(self.folder_add_section, font=self.normal_font, width=50)
self.folder_name_add_entry.pack(fill=X, pady=5)

self.folder_add_btn = Button(self.folder_add_section, text="ð / í .° ì ¶ ¤ ",
                             bg='#333333', fg='white', font=self.normal_font, relief=FLAT, padx=20, pady=8)
self.folder_add_btn.pack(pady=10)

self.folder_add_result = Label(self.folder_add_section, text="", font=self.normal_font,
                               bg='#ffffff', fg='green', wraplength=800, justify=LEFT)
self.folder_add_result.pack(anchor=W, pady=5)

# í .° ì ì ì ¶
self.folder_update_section = Frame(self.folder_frame, bg='#ffffff')
self.folder_update_section.pack(fill=X, pady=10)
self.folder_update_section.pack_forget()

Label(self.folder_update_section, text="¤,°ìí .° ì .°f :", font=self.normal_font, bg='#ffffff')
self.folder_name_old_entry = Entry(self.folder_update_section, font=self.normal_font, width=50)
self.folder_name_old_entry.pack(fill=X, pady=5)

Label(self.folder_update_section, text="ì í .° ì .°f :", font=self.normal_font, bg='#ffffff')
self.folder_name_new_entry = Entry(self.folder_update_section, font=self.normal_font, width=50)
self.folder_name_new_entry.pack(fill=X, pady=5)

self.folder_update_btn = Button(self.folder_update_section, text="ð / í .° ì ì ",
                                bg='#333333', fg='white', font=self.normal_font, relief=FLAT, padx=20, pady=8)
self.folder_update_btn.pack(pady=10)

self.folder_update_result = Label(self.folder_update_section, text="", font=self.normal_font,
                                  bg='#ffffff', fg='green', wraplength=800, justify=LEFT)
self.folder_update_result.pack(anchor=W, pady=5)

# í .° ì >ì ì ¶
self.folder_delete_section = Frame(self.folder_frame, bg='#ffffff')
self.folder_delete_section.pack(fill=X, pady=10)
self.folder_delete_section.pack_forget()

Label(self.folder_delete_section, text="í .° ì .°f :", font=self.normal_font, bg='#ffffff')
self.folder_name_delete_entry = Entry(self.folder_delete_section, font=self.normal_font, width=50)
self.folder_name_delete_entry.pack(fill=X, pady=5)

self.folder_delete_btn = Button(self.folder_delete_section, text="ð / í .° ì >ì ",
                                bg='#f44336', fg='white', font=self.normal_font, relief=FLAT, padx=20, pady=8)
self.folder_delete_btn.pack(pady=10)

self.folder_delete_result = Label(self.folder_delete_section, text="", font=self.normal_font,
                                  bg='#ffffff', fg='green', wraplength=800, justify=LEFT)
self.folder_delete_result.pack(anchor=W, pady=5)

def create_summary_section(self):
    """ì ì %ì °f< ì ¶ """
    self.summary_frame = LabelFrame(self.content_frame, text="ð ì £...ì í /ì ° °° í µì < ì
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        font=self.heading_font, bg='#ffffff', padx=25, pady=25)
    self.summary_frame.pack(fill=BOTH, expand=True, pady=10)
    self.summary_frame.pack_forget()

    Label(self.summary_frame, text="í /ì ° :", font=self.heading_font, bg='#ffffff').pack(anchor=
    self.keyword_label = Label(self.summary_frame, text="",
                                font=self.normal_font, bg='#ffffff', fg='#555', justify=LEFT, and
    self.keyword_label.pack(anchor=W, pady=5, fill=X)

    Label(self.summary_frame, text="ì ì %:", font=self.heading_font, bg='#ffffff').pack(anchor=
    self.summary_label = Label(self.summary_frame,
                                text="",
                                font=self.normal_font, bg='#ffffff', fg='#555', justify=LEFT, and
    self.summary_label.pack(anchor=W, pady=5, fill=X)

    ttk.Separator(self.summary_frame, orient=HORIZONTAL).pack(fill=X, pady=20)

    Label(self.summary_frame, text="ð °‡ í °° ì °‡, í ì /í ,", font=self.heading_font, b
    self.conversion_text = scrolledtext.ScrolledText(self.summary_frame, height=15, width=80,
                                                        font=self.normal_font, wrap=WORD, bg='#ffff
    self.conversion_text.pack(fill=BOTH, expand=True, pady=5)
    self.conversion_text.config(state=DISABLED)

    # ì ì %/í /ì ° í ì /í ,Æ° í ° ì í >ì °§ Æ† °‡.ì .° °ì wraplength ° ì ì ;°ì
    def _update_wrap(event):
        w = event.width - 50 # ì ¢ ì ° ì <°°- °‡.ì (padx 25 * 2)
        if w > 200:
            self.keyword_label.config(wraplength=w)
            self.summary_label.config(wraplength=w)
    self.summary_frame.bind("<Configure>", _update_wrap)

def create_quiz_section(self):
    """í .ì f ì „ì """
    self.quiz_frame = LabelFrame(self.content_frame, text="â ì ì °<,ì ",
                                font=self.heading_font, bg='#ffffff', padx=25, pady=25)
    self.quiz_frame.pack(fill=BOTH, expand=True, pady=10)
    self.quiz_frame.pack_forget()

    # í .ì f í >°""° /
    self.quiz_canvas = Canvas(self.quiz_frame, bg='#ffffff', highlightthickness=0)
    quiz_scrollbar = Scrollbar(self.quiz_frame, orient="vertical", command=self.quiz_canvas.yvie
    self.quiz_content = Frame(self.quiz_canvas, bg='#ffffff')

    self.quiz_canvas.pack(side=LEFT, fill=BOTH, expand=True)
    quiz_scrollbar.pack(side=RIGHT, fill=Y)
    self.quiz_canvas.configure(yscrollcommand=quiz_scrollbar.set)

    # Canvasì ° °„ ì °§ ì ¶° quiz_contentÆ° í ì ¥° ° °ì ì /ì
    self.quiz_window = self.quiz_canvas.create_window((0, 0), window=self.quiz_content, anchor="

    def _update_quiz_canvas(event):
        # Canvas ° °„ ì °§ ì ¶° quiz_content ° °„ ì ;°ì
        canvas_width = event.width
        self.quiz_canvas.itemconfig(self.quiz_window, width=canvas_width)
        self.quiz_canvas.configure(scrollregion=self.quiz_canvas.bbox("all"))

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self.quiz_canvas.bind("<Configure>", _update_quiz_canvas)
self.quiz_content.bind("<Configure>", lambda e: self.quiz_canvas.configure(scrollregion=self

# ° ì í .ì f ì ì fl °f<ì /í ,
self.quiz_frames = []

# ° /ì °<,ì ì ì - °† í ...
self.add_quiz_btn = Button(self.quiz_content, text="â ° /ì °<,ì ì ì -",
    bg='#2196F3', fg='white', font=self.heading_font, relief=FLAT, padx=30, pady=15)
self.add_quiz_btn.pack(pady=10)

# ì - ì °† í ... °° ¶†°¶‡...
ttk.Separator(self.quiz_content, orient=HORIZONTAL).pack(fill=X, pady=20)

self.final_score_label = Label(self.quiz_content, text="", font=self.heading_font,
    bg='ffffff', fg='#333')
self.final_score_label.pack(pady=10)
self.final_score_label.pack_forget()

self.score_btn = Button(self.quiz_content, text="â ì ° µ í ì , °° ì - ì ",
    bg='#333333', fg='white', font=self.heading_font, relief=FLAT, padx=30, pady=15)
self.score_btn.pack(pady=20)
```